

培道中學校

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年級：

科目：

練習題 18.4

$$3. \text{R.H.S} = \frac{1}{2} \int_{\partial \Omega} \cos(\vec{p}, \vec{n}) d\sigma = \frac{1}{2} \int_{\partial \Omega} \frac{\vec{p}}{p} \cdot \vec{n} d\sigma = \frac{1}{2} \int_{\partial \Omega} \nabla \cdot \left(\frac{\vec{p}}{p} \right) d\mu$$

$$= \frac{1}{2} \int_{\partial \Omega} \frac{p - (x-a) \cdot \frac{2(x-a)}{2p}}{p^2} d\mu = \frac{1}{2} \int_{\partial \Omega} \left(\frac{3}{p} - \frac{p^2}{p^3} \right) d\mu = \int_{\partial \Omega} \frac{1}{p} d\mu = \iiint_{\Omega} \frac{1}{p} dx dy dz = \text{L.H.S}$$

4. (1) 記 $\Sigma: x^2+y^2+z^2=a^2$ 被 $x+y+z=0$ 截下的曲面, $\partial \Sigma = \Gamma$

$$\int_{\Gamma} y dx + z dy + x dz = \int_{\Sigma} \begin{vmatrix} \frac{x}{a} & \frac{y}{a} & \frac{z}{a} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & z & x \end{vmatrix} d\sigma = \int_{\Sigma} \left(-\frac{x}{a} + (-\frac{y}{a}) + (-\frac{z}{a}) \right) d\sigma$$

$$\rightarrow \int_{\Sigma} = - \int_{\Sigma} dy dz + dz dx + dx dy$$

記 $\Omega: x^2+y^2+z^2 \leq a^2$ 與 $x+y+z=0$ 圍成的閉區域

$S: x^2+y^2+z^2 \leq a^2$ 且 $x+y+z \geq 0$ 的曲面

$$\partial \Omega = \Sigma \cup S$$

$$\int_{\partial \Omega} dy dz + dz dx + dx dy = \int_{\partial \Omega} 0 d\mu = 0 = \int_{\Sigma} + \int_{S}$$

$$-\int_{\Sigma} = \int_S (1, 1, 1) \cdot \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right) d\sigma = -\sqrt{3} \int_S d\sigma = -\sqrt{3} \pi a^2$$

(2) 記 $\Sigma: x^2+y^2=2y$ 被 $y=z$ 所截下的曲面, $\partial \Sigma = \Gamma + \{(x, y, z) | z=0, x^2+(y-1)^2=1\}$

$$\int_{\Gamma} = \int_{\Sigma} \begin{vmatrix} x & y-1 & 0 \\ D_x & D_y & D_z \\ y+z & z+x & x+y \end{vmatrix} d\sigma = 0$$

$$\int_{\Gamma} = - \int_{\Gamma} y dx + x dy$$

(2) 記 $\Sigma: x^2 + (y-1)^2 \leq 1 \cap y=z$, $\vec{n} = \frac{1}{\sqrt{2}}(0, -1, 1)$, $\partial\Sigma = \Gamma$

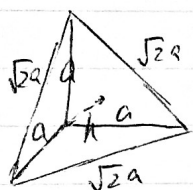
$$\int_{\Gamma} = \frac{1}{\sqrt{2}} \int_{\Sigma} \begin{vmatrix} 0 & -1 & 1 \\ D_x & D_y & D_z \\ y+z & 2x & x+y \end{vmatrix} d\sigma = 0$$

(3) 記 $\Sigma: x^2 + y^2 + z^2 \leq a^2$ 被 $x+y+z=a$ 截下的曲面, $\Gamma = \partial\Sigma$, $\vec{n} = \frac{1}{a}(x, y, z)$

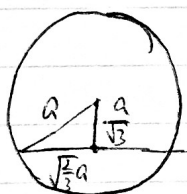
$$\int_{\Gamma} = \frac{1}{a} \int_{\Sigma} \begin{vmatrix} x & y & z \\ D_x & D_y & D_z \\ y^2 & z^2 & x^2 \end{vmatrix} d\sigma = \frac{1}{a} \int_{\Sigma} (-2zx - 2xy - 2yz) d\sigma$$

(3) 記 $\Sigma: \{(x, y, z) \mid x^2 + y^2 + z^2 \leq a^2 \cap x+y+z=a\}$, $\partial\Sigma = \Gamma$, $\vec{n} = \frac{1}{\sqrt{3}}(1, 1, 1)$

$$\int_{\Gamma} = \frac{1}{\sqrt{3}} \int_{\Sigma} \begin{vmatrix} D_x & D_y & D_z \\ y^2 & z^2 & x^2 \end{vmatrix} d\sigma = \frac{1}{\sqrt{3}} \int_{\Sigma} -2(z+x+y) d\sigma = -\frac{2a}{\sqrt{3}} \int_{\Sigma} d\sigma$$



$$\frac{1}{6}a^3 \cdot \frac{1}{\beta} = \frac{1}{6} \cdot 2a^2 \cdot \sin 60^\circ \cdot h \cdot \frac{1}{\beta} \Rightarrow h = \frac{a}{\sqrt{3}}$$



$$\int_{\Sigma} d\sigma = \pi \cdot \left(\frac{\sqrt{2}}{3}a\right)^2 = \frac{2}{3}\pi a^2 \Rightarrow \int_{\Gamma} = -\frac{4}{3\sqrt{3}}\pi a^3$$

5. 記 $\vec{p} = (x, y, z)$, $\vec{a} = (r, s, t)$

$$\text{L.H.S.} = \int_{\partial\Sigma} \vec{a} \times \vec{p} \cdot d\vec{p} = \int_{\partial\Sigma} \begin{vmatrix} dx & dy & dz \\ r & s & t \\ x & y & z \end{vmatrix} = \int_{\Sigma} \begin{vmatrix} \cos\alpha & \cos\beta & \cos\gamma \\ D_x & D_y & D_z \\ \begin{vmatrix} s & t \\ y & z \end{vmatrix} & \begin{vmatrix} t & r \\ z & x \end{vmatrix} & \begin{vmatrix} r & s \\ x & y \end{vmatrix} \end{vmatrix} d\sigma$$

$$= \int_{\Sigma} (D_y \begin{vmatrix} r & s \\ x & y \end{vmatrix} - D_z \begin{vmatrix} t & r \\ z & x \end{vmatrix}, D_z \begin{vmatrix} s & t \\ y & z \end{vmatrix} - D_x \begin{vmatrix} r & s \\ x & y \end{vmatrix}, D_x \begin{vmatrix} t & r \\ z & x \end{vmatrix} - D_y \begin{vmatrix} s & t \\ y & z \end{vmatrix}) \cdot \vec{n} \cdot d\sigma$$

$$= \int_{\Sigma} (r - (-r), s - (-s), t - (-t)) \cdot \vec{n} \cdot d\sigma = 2 \int_{\Sigma} \vec{a} \cdot \vec{n} \cdot d\sigma$$

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練習題 18.5

$$1. (1) (x dx + y dy) \wedge (z dz - z dx) = xz dx \wedge dz + yz dy \wedge dz - yz dy \wedge dx \\ = yz dy \wedge dz - zx dz \wedge dx + yz dx \wedge dz$$

$$(2) (dx + dy + dz) \wedge (x dx \wedge dy - z dy \wedge dz) \\ = (-z+x) dx \wedge dy \wedge dz = (x-z) dx \wedge dy \wedge dz$$

$$2. (1) dw = (y+z) dx + (z+x) dy + (x+y) dz$$

$$(2) dw = (y dx + x dy) \wedge dx = -x dx \wedge dy$$

$$(3) dw = (y dx + x dy + z dy + y dz) \wedge dx = y dz \wedge dx - (z+x) dx \wedge dy$$

$$(4) dw = (y dx + x dy) \wedge dx + x^2 (z dx \wedge dy) = x dx \wedge dy$$

$$(5) dw = (2xy dx + x^2 dy) \wedge dx - (yze^x dx + ze^x dy + ye^x dz) \wedge dy \\ = -x^2 dx \wedge dy - yze^x dx \wedge dy - e^x y dy \wedge dz \\ = -(x^2 + e^x y z) dx \wedge dy - e^x y dy \wedge dz$$

$$(6) dw = (y^2 dx + 2xy dy) \wedge dy \wedge dz - (z^2 dx + 2xz dz) \wedge dx \wedge dy \\ = (y^2 - 2xz) dz \wedge dy \wedge dx$$

$$(7) dw = (y+z+x) dx \wedge dy \wedge dz$$

$$3. w = \sum_{i_1, i_2, \dots, i_p} a_{i_1, i_2, \dots, i_p}(x_1, x_2, \dots, x_n) dx_{i_1} \wedge dx_{i_2} \wedge \dots \wedge dx_{i_p}$$

$$dw = \sum_{\{i_1, i_2, \dots, i_p\} \cap \{j_1, j_2, \dots, j_p\} = \emptyset} a_{i_1, i_2, \dots, i_p} dx_{i_1} \wedge \dots \wedge dx_{i_p} \wedge dx_{j_1} \wedge \dots \wedge dx_{j_p}$$

$$dw = \sum_{i_1, i_2, \dots, i_p} \sum_{\substack{k=1 \\ j_k \notin \{i_1, i_2, \dots, i_p\}}}^{n-p} \frac{\partial a_{i_1, i_2, \dots, i_p}}{\partial x_{j_k}} dx_{j_k} \wedge dx_{i_1} \wedge \dots \wedge dx_{i_p}$$

$$d(dw) = \sum_{i_1, i_2, \dots, i_p} \sum_{\substack{k=1 \\ j_k \notin \{i_1, i_2, \dots, i_p\}}}^{n-p} \sum_{\substack{l=1 \\ j_l \notin \{i_1, i_2, \dots, i_p\}}}^{n-p} \frac{\partial a_{i_1, i_2, \dots, i_p}}{\partial x_{j_k} \partial x_{j_l}} dx_{j_l} \wedge dx_{j_k} \wedge dx_{i_1} \wedge \dots \wedge dx_{i_p}$$

$$d(dw) = \sum_{i_1, i_2, \dots, i_p} \sum_{\substack{k=1 \\ j_k \notin \{i_1, i_2, \dots, i_p\}}}^{n-p} \sum_{\substack{l=1 \\ j_l \notin \{i_1, i_2, \dots, i_p\}}}^{n-p} \frac{\partial a_{i_1, i_2, \dots, i_p}}{\partial x_{j_k} \partial x_{j_l}} dx_{j_l} \wedge dx_{j_k} \wedge dx_{i_1} \wedge \dots \wedge dx_{i_p}$$

$$\text{因為 } w \in C^2, \text{ 所以 } \frac{\partial a_{i_1, i_2, \dots, i_p}}{\partial x_{j_k} \partial x_{j_l}} = \frac{\partial a_{i_1, i_2, \dots, i_p}}{\partial x_{j_l} \partial x_{j_k}}$$

$$\text{同時有 } x_{j_k} \wedge x_{j_l} = -x_{j_l} \wedge x_{j_k} \Rightarrow d(dw) = 0$$

練習題 19.1

1. 設 $f, g: \Omega \rightarrow \mathbb{R}$, $\Omega \subset \mathbb{R}^n$

$$\nabla\left(\frac{f}{g}\right) = \left[\frac{\partial(\frac{f}{g})}{\partial x_i} \right]_{1 \times n} = \left[\frac{\frac{\partial f}{\partial x_i} g - f \frac{\partial g}{\partial x_i}}{g^2} \right]_{1 \times n} = \frac{1}{g^2} \left(\left[\frac{\partial f}{\partial x_i} \right]_{1 \times n} g - f \left[\frac{\partial g}{\partial x_i} \right]_{1 \times n} \right)$$

$$= \frac{1}{g^2} (\cancel{g} \nabla f - f \nabla g)$$

2. 設 ~~$u: \Omega \rightarrow \mathbb{R}^k$~~ $u: \Omega \rightarrow \mathbb{R}$, $\Omega \subset \mathbb{R}^n$, $\vec{f}: \Omega \rightarrow \mathbb{R}^m$, $\vec{f} = (f_1, f_2, \dots, f_m)$

$$\nabla(u \cdot \vec{f}) = \nabla(u f_1, u f_2, \dots, u f_m)$$

$$= \left[\frac{\partial u}{\partial x_i} f_j + u \frac{\partial f_j}{\partial x_i} \right]_{n \times m} = [\nabla u]_{n \times 1}^T \vec{f} + u J(\vec{f})$$

3. (1) $\nabla \log p = \left(\frac{\partial \log p}{\partial p} \cdot \frac{\partial p}{\partial x}, \frac{\partial \log p}{\partial p} \cdot \frac{\partial p}{\partial y}, \frac{\partial \log p}{\partial p} \cdot \frac{\partial p}{\partial z} \right) = \frac{1}{p^2} (x, y, z) = \frac{\vec{p}}{p^2}$

(2) $\nabla f(p) = \frac{\partial f}{\partial p} \nabla p = \frac{\partial f}{\partial p} \cdot \frac{\vec{p}}{p}$ (3) $\nabla f(p^2) = \frac{\partial f(p^2)}{\partial t} \cdot \nabla p^2 = 2 \cdot \frac{\partial f(p^2)}{\partial t} \cdot \vec{p}$

(4) ~~$\nabla(f(p) \vec{p} \cdot \vec{a}) = f \vec{p} \cdot \vec{a} \cdot \frac{\partial f}{\partial p}$~~

(4) $\nabla(f(p) \cdot \vec{p} \cdot \vec{a}) = \vec{p} \cdot \vec{a} \cdot \nabla f(p) + f(p) \nabla(\vec{p} \cdot \vec{a})$
 $= (\vec{p} \cdot \vec{a}) \cdot \frac{\partial f}{\partial p} \cdot \frac{\vec{p}}{p} + f(p) \vec{a}$

4. $\frac{\partial f}{\partial(\frac{\nabla g}{\|\nabla g\|})} = \nabla f \cdot \nabla g \cdot \frac{1}{\|\nabla g\|} = \frac{1}{\|\nabla g\|} \nabla f \cdot \nabla g$. 當且僅當 $\nabla f \cdot \nabla g = 0 \Leftrightarrow \nabla f \perp \nabla g$ 時
 變化率 = 0

5. 只需證 $\frac{\partial u}{\partial x}(\vec{p}) = \lim_{\Omega \rightarrow \vec{p}} \frac{1}{\mu(\Omega)} \iint_{\Omega} u \, dy \, dz$

由 Gauss 公式知, $\iint_{\partial \Omega} u \, dy \, dz = \int_{\Omega} \frac{\partial u}{\partial x} \, d\mu$, 由積分中值定理知, $\exists \xi \in \Omega$

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練習題 19.2.

$$1. \text{ 當 } p > 0 \text{ 時, } \Delta \log p = \nabla \cdot \nabla \log p = \nabla \cdot \left(\frac{1}{p} \nabla p \right) = \nabla \cdot \left(\frac{1}{p^2} \vec{p} \right) \\ = \frac{1}{p^2} \nabla \cdot \vec{p} + \vec{p} \cdot \nabla \frac{1}{p^2} = \frac{2}{p^2} + \vec{p} \cdot \left(-\frac{2}{p^3} \cdot \frac{\vec{p}}{p} \right) = 0$$

$$2. \Delta(fg) = \nabla \cdot \nabla(fg) = \nabla \cdot (g \nabla f + f \nabla g) = \nabla \cdot (g \nabla f) + \nabla \cdot (f \nabla g) \\ = g \Delta f + \nabla f \cdot \nabla g + f \Delta g + \nabla g \cdot \nabla f = f \Delta g + g \Delta f + 2 \nabla f \cdot \nabla g$$

$$3. (1) \int_{\partial \Omega} \frac{\partial u}{\partial \vec{n}} d\sigma = \int_{\partial \Omega} \nabla u \cdot \vec{n} d\sigma = \int_{\Omega} \nabla \cdot \nabla u d\mu = \int_{\Omega} \Delta u d\mu \quad (\text{Gauss 公式})$$

$$(2) \int_{\partial \Omega} v \frac{\partial u}{\partial \vec{n}} d\sigma = \int_{\partial \Omega} v \nabla u \cdot \vec{n} d\sigma = \int_{\Omega} \nabla \cdot (v \nabla u) d\mu = \int_{\Omega} (v \Delta u + \nabla u \cdot \nabla v) d\mu \quad (\text{Gauss 公式})$$

$$(3) \int_{\partial \Omega} \left| \frac{\frac{\partial u}{\partial \vec{n}}}{u} \frac{\frac{\partial v}{\partial \vec{n}}}{v} \right| d\sigma = \int_{\partial \Omega} v \frac{\partial u}{\partial \vec{n}} d\sigma - \int_{\partial \Omega} u \frac{\partial v}{\partial \vec{n}} d\sigma = \int_{\Omega} (v \Delta u + \nabla u \cdot \nabla v - u \Delta v - \nabla v \cdot \nabla u) d\mu \\ = \int_{\Omega} \left| \frac{\Delta u}{u} - \frac{\Delta v}{v} \right| d\mu \quad (\text{利用 (2) 的結論})$$

$$4. (1) \int_{\partial \Omega} \frac{\partial u}{\partial \vec{n}} d\sigma = \int_{\partial \Omega} \nabla u \cdot \vec{n} d\sigma = \int_{\Omega} \Delta u d\mu = \int_{\Omega} 0 d\mu = 0$$

$$(2) \int_{\partial \Omega} u \frac{\partial u}{\partial \vec{n}} d\sigma = \int_{\Omega} (u \Delta u + \nabla u \cdot \nabla u) d\mu = \int_{\Omega} (\nabla u)^2 d\mu = \int_{\Omega} \|\nabla u\|^2 d\mu$$

練習題 19.3

$$1. \nabla \times (\nabla \times \vec{F}) = \nabla \times \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ D_x & D_y & D_z \\ P & Q & R \end{vmatrix} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ D_x & D_y & D_z \\ |D_y & D_z| & |D_z & D_x| & |D_x & D_y| \\ P & Q & R \end{vmatrix}$$

對於第一個分量 $\frac{\partial^2 Q}{\partial y \partial x} - \frac{\partial^2 P}{\partial y^2} - \frac{\partial^2 P}{\partial z^2} + \frac{\partial^2 R}{\partial z \partial x}$

$$\frac{\partial^2 Q}{\partial y \partial x} + \frac{\partial^2 R}{\partial z \partial x} + \frac{\partial^2 P}{\partial x^2} - \left(\frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2} + \frac{\partial^2 P}{\partial z^2} \right)$$

$$= \frac{\partial}{\partial x} (\nabla \cdot \vec{F}) - \nabla^2 P$$

$$\nabla \times (\nabla \times \vec{F}) = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) (\nabla \cdot \vec{F}) - \nabla^2 (P, Q, R) = \nabla (\nabla \cdot \vec{F}) - \nabla^2 \vec{F}$$

$$2. (1) \vec{n} \times \vec{F} d\sigma = -\vec{F} \times \vec{n} d\sigma = - \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ P & Q & R \\ dy dz & dz dx & dx dy \end{vmatrix}$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ dy dz & dz dx & dx dy \\ P & Q & R \end{vmatrix} =$$

對於第一個分量 $R dz dx - Q dx dy$

$$\iint_{\partial \Omega} R dz dx - Q dx dy = \iiint_{\Omega} \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) dx dy dz = \left(\frac{\partial R(\vec{x})}{\partial y} - \frac{\partial Q(\vec{x})}{\partial z} \right) \mu(\Omega)$$

所以有 $\int_{\partial \Omega} \vec{n} \times \vec{F} d\sigma = \mu(\Omega) \left(\frac{\partial R(\vec{x})}{\partial y} - \frac{\partial Q(\vec{x})}{\partial z}, \frac{\partial P(\vec{y})}{\partial z} - \frac{\partial R(\vec{y})}{\partial x}, \frac{\partial Q(\vec{z})}{\partial x} - \frac{\partial P(\vec{z})}{\partial y} \right)$

所以 $\lim_{\Omega \rightarrow \vec{p}} \frac{1}{\mu(\Omega)} \int_{\partial \Omega} \vec{n} \times \vec{F} d\sigma = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) (\vec{p}) = \text{rot } \vec{F}(\vec{p})$

(2) $\text{div}(f \text{grad } f) = f \Delta f + \|\nabla f\|^2 = bf + f \Delta f = af \Rightarrow f \Delta f = (a-b)f \Rightarrow \Delta f = a-b$

$$\int_{\partial \Omega} \frac{\partial f}{\partial \vec{n}} d\sigma = \int_{\partial \Omega} \nabla f \cdot \vec{n} d\sigma = \int_{\Omega} \Delta f d\mu = (a-b) \mu(\Omega)$$